

SARVESH BASKAR

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PROFESSIONAL SUMMARY

Research assistant working on multimodal AI, vision-language models, planning, and reliable agentic systems. Research experience with UMD and UMBC, with publications in LLM personalization, VLM hallucination mitigation, video reasoning, code reasoning, and multimodal evaluation.

EDUCATION

Birla Institute of Technology and Science (BITS) Pilani

Sep 2020 to July 2025

Dual Degree:

Bachelor of Engineering in Computer Science

Cumulative GPA: 8.46/10.00

Master of Science in Physics

Thesis Title: An Approach to Resolving the Persona Knowledge Gap in LLMs during Multi-Turn Conversations

SKILLS

Programming: Python, C/C++, Bash, SQL, LaTeX

ML/AI: PyTorch, TensorFlow, OpenCV, Transformers, vLLM, LangChain, DSPy, RAG, VLMs

Systems: Git, Linux, Docker, Slurm, Redis, ChromaDB, FAISS, Ollama, MCP

PUBLICATIONS

- Agrawal, Aakriti, Gouthaman KV, Rohith Aralikatti, Gauri Jagatap, Jiaxin Yuan, **Sarvesh Baskar**, Vijay Kamarshi, Andrea Fanelli, and Furong Huang. “Towards Mitigating Hallucinations in Large Vision-Language Models by Refining Textual Embeddings.” *ACL Findings*, 2026.
- **Baskar, Sarvesh**, Muhammad R. Islam, Zikui Cai, Anirudh Satheesh, Tom Goldstein, and Furong Huang. “Parametric Capability Profiling for Multimodal Video Reasoning.” *Under Review ECCV*, 2026.
- Islam, Muhammad R., **Sarvesh Baskar**, Zikui Cai, Anirudh Satheesh, Keenan Powell, Tom Goldstein, and Furong Huang. “Executable Reasoning Traces for Diagnosing Multimodal Video Reasoning.” *Under Review ECCV*, 2026.
- **Baskar, Sarvesh**, Muhammad R. Islam, Zikui Cai, Ankit Nakhawa, Anirudh Satheesh, Tom Goldstein, and Furong Huang. “The Low-Frequency Trap: Why Scaling Doesn’t Solve Simple Temporal Counting.” *ICLR 2026 Workshop on I Can’t Believe It’s Not Better*, 2026.
- Mohseni, Seyedreza, **Sarvesh Baskar**, Arman Hossain, Aritran Piplai, Edward Raff, and Manas Gaur. “Analyzing Chain of Thought (CoT) Approaches in Control Flow Code Deobfuscation Tasks.” *WoRMA*, 2026.
- **Baskar, Sarvesh**, Manas Gaur, Srinivasan Parthasarathy, and Tanmay Tulsidas Verlekar. “(CPER) From Guessing to Asking: An Approach to Resolving Persona Knowledge Gap in LLMs during Multi-Turn Conversations.” *NAACL SRW*, 2025.

EXPERIENCE

AI Research Assistant

Jul 2025 - Present

University of Maryland, College Park

Remote

- Built and evaluated multimodal AI pipelines for large vision-language models, video reasoning, hallucination mitigation, and synthetic visual reasoning tasks.

- Implemented embedding-refinement experiments in OpenQwen2VL to improve visual grounding and reduce hallucinations in large vision-language models.
- Ran controlled experiments, ablations, result analysis, and paper-ready visualizations for ACL and ECCV research submissions.

AI Consultant

Aug 2025 - Present

Avyott

Remote

- Designed and deployed multimodal RAG chatbots for enterprise clients, supporting grounded Q&A over text, images, tables, and internal documents.
- Built MCP servers to integrate chat workflows with ticketing systems, enabling ticket creation, updates, and status tracking through natural language.
- Improved user intent capture by adding clarification workflows inspired by multi-turn LLM research, making responses more personalized and reliable.

AI Research Assistant

Aug 2024 - Jul 2025

University of Maryland, Baltimore County

MD, USA

- Developed LLM-based systems for multi-turn conversation, planning, and code reasoning using Python, prompt engineering, and evaluation pipelines.
- Led experiments and analysis for a NAACL 2025 paper on persona knowledge gaps in LLM conversations, improving response relevance in long dialogues.
- Contributed to IBM-collaborative planning research by building benchmarks, experiment logs, and evaluation scripts for natural-language planning tasks.

AI / Software Intern

May 2024 - Jul 2024

Techisy

Remote

- Built an AI-powered duplicate bill detection system using OCR, OpenAI embeddings, and CLIP, achieving 95% accuracy.
- Developed a RAG-based resume matching system using LangChain, Ollama, and FAISS to align resumes with job descriptions.
- Deployed working AI prototypes and iterated with stakeholders to improve screening efficiency and reduce manual review effort.